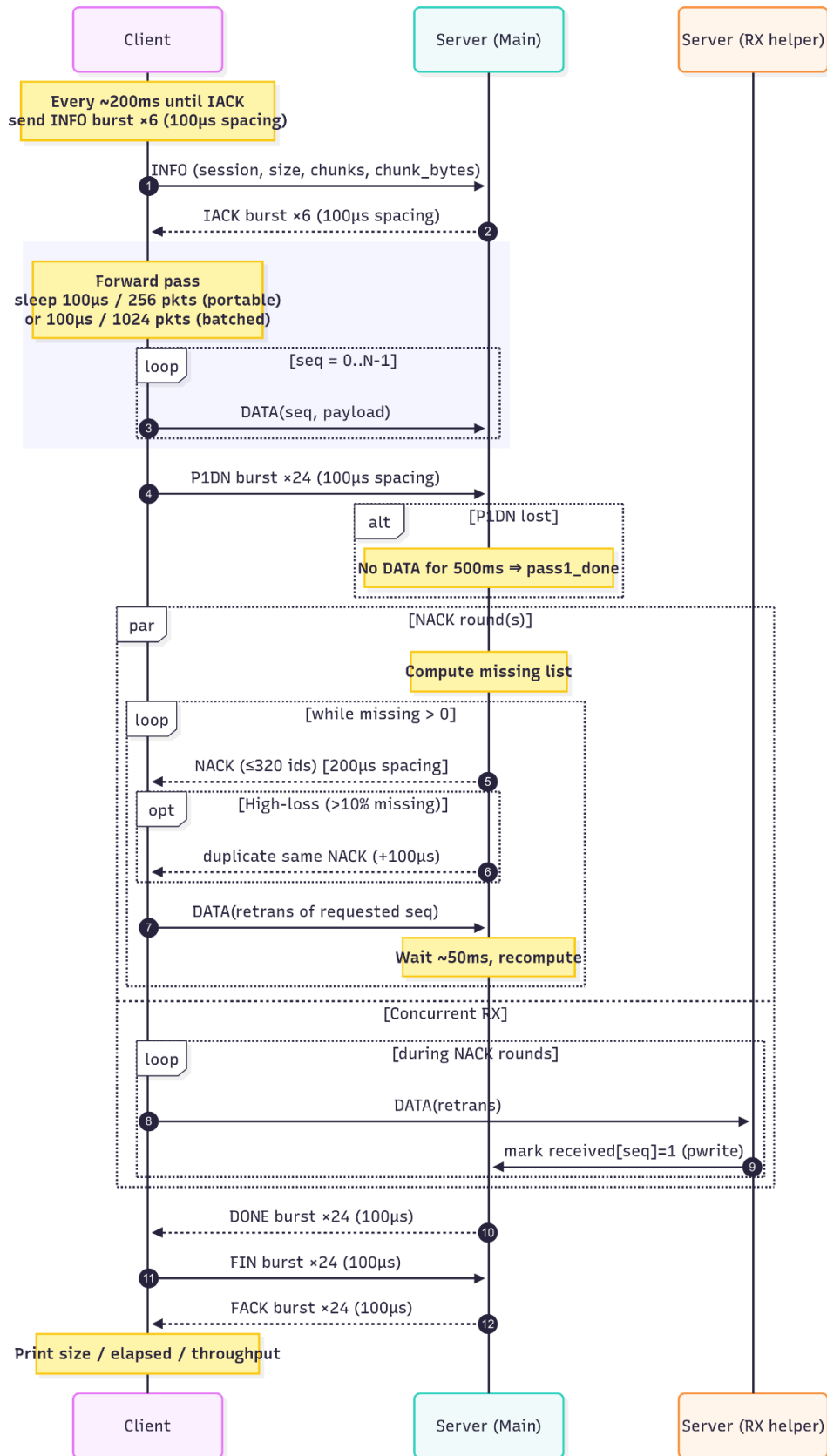




Binaries & roles

- `ft_client`: reads the input file (RAM or mmap), splits into fixed-size chunks, sends a one-pass stream of DATA, then **only** retransmits chunks the server NACKs. Prints size/time/throughput.
- `ft_server`: opens/truncates the output file, writes chunks directly at offsets, computes which chunks are missing, drives **batched NACK** rounds, and finalizes with FIN/FACK.

Wire messages (4-byte tags)

- INFO → client announces {chunk_bytes, file_size, chunk_count, session_id}
- IACK ← server acknowledges session (now sent as a **small burst** for robustness)
- DATA → {seq, payload_len} + <payload>
- P1DN → end of first pass
- NACK ← batched list of missing seq IDs (≤ 320 /packet)
- DONE ← server has no more gaps
- FIN / FACK →/← teardown

Key constants (from code)

- DEFAULT_CHUNK=1460 (choose ~8960 for jumbo, clamp ≤ 65535)
- MAX_NACK_IDS_PER_MSG=320
- INFO_REPEATS=6 (both INFO burst on client and IACK burst on server)
- CTRL_REPEATS=24 (P1DN/DONE/FIN/FACK bursts)
- Socket buffers: **16 MB** send & receive
- Poll timeouts: **200 ms**
- Pass-1 idle cutoff on server: **500 ms** (if P1DN is lost)
- NACK inter-packet spacing: **200 μ s**
- NACK round recompute delay: **50 ms** (new)

- High-loss mode (server): if >**10%** chunks missing, **repeat each NACK batch once** with **100 µs** gap (new)

Concurrency

- **Client**: single send thread; optional ERRQUEUE drainer thread when SO_ZEROCOPY is enabled.
- **Server**: main thread + a **nonblocking RX helper thread** during NACK rounds so retransmitted DATA arrive while NACKs are being sent.

MTU & sizing

- Choose --chunk so $\text{sizeof(DataMsg)} + \text{chunk} + \text{UDP+IP} \leq \text{MTU}$.
MTU 1500 → --chunk 1460. Jumbo ≈9000 → --chunk ~8960.

Performance knobs

- -batch N (Linux): sendmmsg to cut syscalls (typ. 16–64).
- -zerocopy (Linux): SO_ZEROCOPY with background ERRQUEUE drain.
- -rate-mbit X: kernel pacing via SO_MAX_PACING_RATE (use fq qdisc).
- -mmap: zero extra user-space copy for file reads.

Code Explanation (step-by-step flow with exact intervals)

1) Handshake (loss-tolerant)

1. **Client** sends **INFO bursts** every **~200 ms**: inside each burst, **INFO_REPEATS=6** copies spaced **100 µs** apart.
2. **Server** on first valid INFO:
 - Stores session_id, clamps chunk_bytes ≤ 65535, sizes bookkeeping, opens & best-effort ftruncate(out, file_size).
 - Replies with an **IACK burst** of **6 copies** (INFO_REPEATS), **100 µs** apart.
 - If a duplicate INFO arrives later (client missed IACK), server **re-sends the IACK burst** immediately.

3. **Client** polls up to **300 ms** per loop iteration and proceeds once a matching IACK(session_id) is received.

Why bursts? Early loss/congestion shouldn't stall session start.

2) Forward pass (fast, one shot)

- **Client** sends DATA(seq, payload) for seq=0..chunk_count-1. Payload is chunk_bytes (shorter only for final chunk).
- **Pacing while sending:**
 - **Portable path (send/sendto):** sleep **100 µs every 256 packets**.
 - **Linux fast path** (when --batch or --zerocopy or --mmap is used):
 - Build mmsghdr arrays and call **sendmmsg**; optional MSG_ZEROCOPY.
 - Background **ERRQUEUE drainer thread** if zerocopy is on.
 - Tiny sleep **100 µs every 1024 packets** to avoid burst loss.
 - Optional **kernel pacing** at --rate-mbit X.
- **Server pass-1 receive** (recv_loop):
 - Polls with **200 ms** timeout; for each DATA, validates, then **pwrite to offset = seq * chunk_bytes** and marks received[seq]=1.
 - Pass-1 ends when **P1DN** arrives **or** after **500 ms** with no DATA (handles lost P1DN).
- **Client** sends **P1DN burst** of **24 copies** (CTRL_REPEATS), **100 µs** apart.

3) Gap recovery (batched NACK rounds)

1. **Server** scans received[] (0/1 per chunk) → builds missing list.
2. **Server sends NACKs in batches:**
 - ≤ **320 seq IDs** per NACK to stay under MTU (MAX_NACK_IDS_PER_MSG).
 - **200 µs** pause between NACK packets.
 - **High-loss mode** (new): if currently missing > **10%** of chunks, **repeat each NACK packet once** with an extra **100 µs** gap.

3. **Client**, on each NACK, **re-sends only the requested seq IDs** (same send path & pacing as pass-1).
4. **Server** keeps receiving retransmitted DATA **concurrently** in the RX helper thread.
5. After sending a full NACK sweep, **server waits ~50 ms** (new), rescans, and repeats steps 2–4 until no chunks are missing.
6. When complete, **server sends DONE burst of 24 copies** (100 μ s apart).

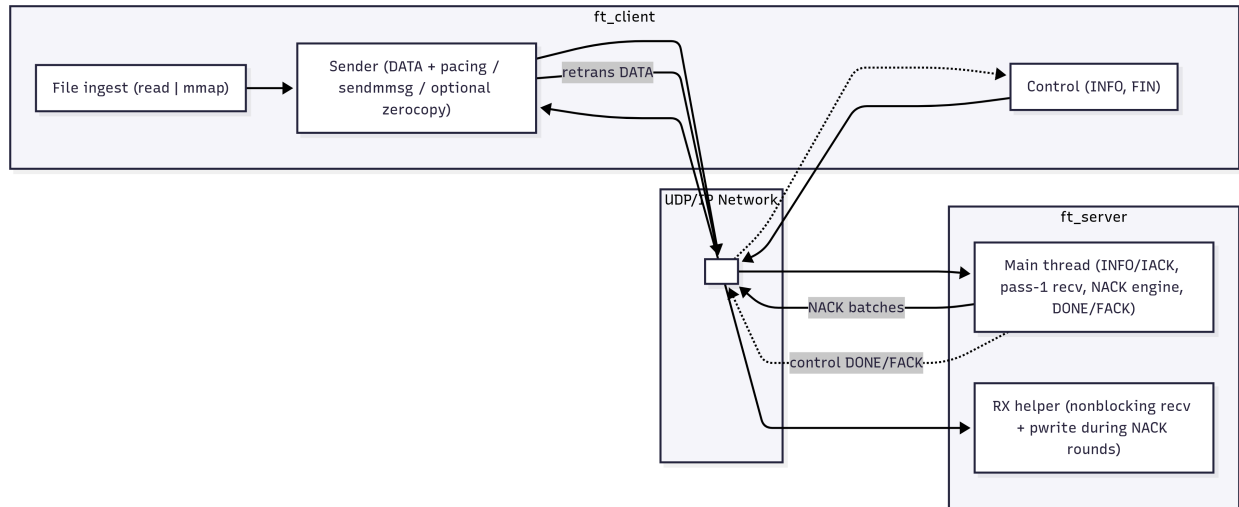
Intervals to remember

- NACK packet spacing: **200 μ s**
- Extra duplicate NACK in high loss: **+100 μ s** between duplicates
- Recompute delay between rounds: **~50 ms**
- Client pacing: **100 μ s / 256 pkts** (portable) or **100 μ s / 1024 pkts** (batched)

4) Teardown & timing

1. **Client**, on DONE, sends **FIN burst of 24 copies** (100 μ s apart).
2. **Server** replies with **FAck burst of 24 copies**, then fsync + close.
3. **Client** prints:
 - Transfer size (Mbits)
 - Elapsed time (sec)
 - Throughput (Mbit/s)

Timing window in code: from before first INFO to after FACK. If you need data-only timing, start at first DATA send and stop when DONE is received.



5) Error-handling & safety (built-in)

- **All control** (IACK, P1DN, DONE, FIN/FACK) is sent in **bursts** to tolerate loss.
- **Duplicate INFO** is handled: server re-issues IACK bursts.
- **In-place writes** + 1-byte presence map → no reassembly buffer; duplicates harmless.
- **Large socket buffers** (16 MB) cushion high-BDP paths.
- Single session per server process (NACK doesn't carry session_id—OK for the lab setup).

6) What's changed vs. earlier version (the subtle bits)

- **IACK is now a burst** (not single shot) and **duplicate INFO triggers IACK re-send**.
- **NACK rounds are tighter**: recompute after **~50 ms** (was ~100 ms).
- **High-loss assist**: **duplicate each NACK batch** when >10% chunks missing.

TL;DR (one screen)

1. **INFO×6 every ~200 ms ↔ IACK×6**
2. **DATA 0..N-1** with light pacing → **P1DN×24**
3. Server: find gaps → **NACK(batch ≤320)** with **200 μs** spacing

- If high loss: send each NACK **twice** (+100 μ s)
- Wait **~50 ms**, rescan, repeat

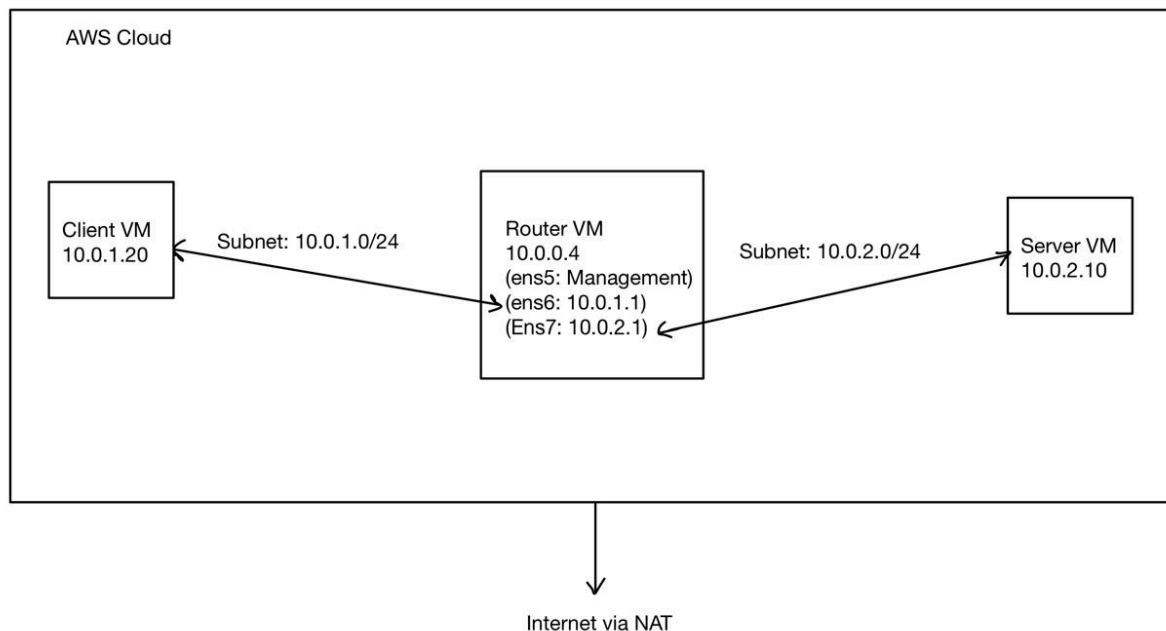
4. **DONE×24** $\leftarrow$ server $\rightarrow$ **FIN×24** $\rightarrow$ client $\leftarrow$ **FAK×24**

5. Client prints size / time / throughput

Network Topology

We will be using the network topology created in AWS

Description for the lab environment is provided below



The above diagram shows the topology we used to conduct the experiment

VPC: Custom Lab2-VPC (10.0.0.0/16), with dedicated subnets for client, server, router management and private subnets (for client, server, and router) are

associated with a route to the IGW, allowing outbound internet connectivity as needed.

Virtual private cloud

Your VPCs (1/1)

Find VPCs by attribute or tag

VPC ID: vpc-047a81acb2d7317b9

Clear filters

Name	VPC ID	State	Block Public...	IPv4 CIDR	IPv6 CIDR	DHCP option set	Main route table
Lab2-VPC	vpc-047a81acb2d7317b9	Available	Off	10.0.0.0/16	-	dopt-03385a3bdc1cf11...	rtb-0705e51d1ea73688d

vpc-047a81acb2d7317b9 / Lab2-VPC

Details | **Resource map** | CIDRs | Flow logs | Tags | Integrations

Resource map

VPC: Your AWS virtual network

Subnets (3): Subnets within this VPC

- us-west-2a
 - Router-Mgmt-Subnet
 - Client-Subnet
 - Server-Subnet

Route tables (4): Route network traffic to resources

- Server-RT
- default
- Router-RT
- Client-RT

Network Connections (1): Connections to other networks

- Lab2-IGW

Virtual private cloud

Route tables (4)

Find route tables by attribute or tag

VPC: vpc-047a81acb2d7317b9

Clear filters

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
Server-RT	rtb-05b4a6b935fbc00ed	subnet-0b202a941628e1...	-	No	vpc-047a81acb2d7317b9 Lab...	78353413...
default	rtb-0705e51d1ea73688d	-	-	Yes	vpc-047a81acb2d7317b9 Lab...	78353413...
Router-RT	rtb-0977e61a185316661	subnet-021e24794e7124...	-	No	vpc-047a81acb2d7317b9 Lab...	78353413...
Client-RT	rtb-0fe0fde1dab49d025	subnet-097bd56256b948...	-	No	vpc-047a81acb2d7317b9 Lab...	78353413...

Select a route table

VPC > Internet gateways

VPC dashboard

EC2 Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers **New**

Security

Network ACLs

Security groups

Internet gateways (1/1) Info

Find Internet gateways by attribute or tag

VPC ID : vpc-047a81acb2d7317b9 Clear filters

1

☒	Name	Internet gateway ID	State	VPC ID	Owner
☒	Lab2-IGW	igw-0f9ba3abb216fce0b	Attached	vpc-047a81acb2d7317b9 Lab2-VPC	783534138674

igw-0f9ba3abb216fce0b / Lab2-IGW

Details Tags

Details

Internet gateway ID	State	VPC ID	Owner
igw-0f9ba3abb216fce0b	Attached	vpc-047a81acb2d7317b9 Lab2-VPC	783534138674

EC2 Instances:

- Client VM (10.0.1.20, subnet 10.0.1.0/24)
- Server VM (10.0.2.10, subnet 10.0.2.0/24)
- Router VM (10.0.0.4 with interfaces on multiple subnets: management, client, and server subnets)

EC2 > Instances

EC2

Dashboard

EC2 Global View

Events

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

Instances (5) Info

Find Instance by attribute or tag (case-sensitive)

Running

Last updated 38 minutes ago

Connect

Instance state

Actions

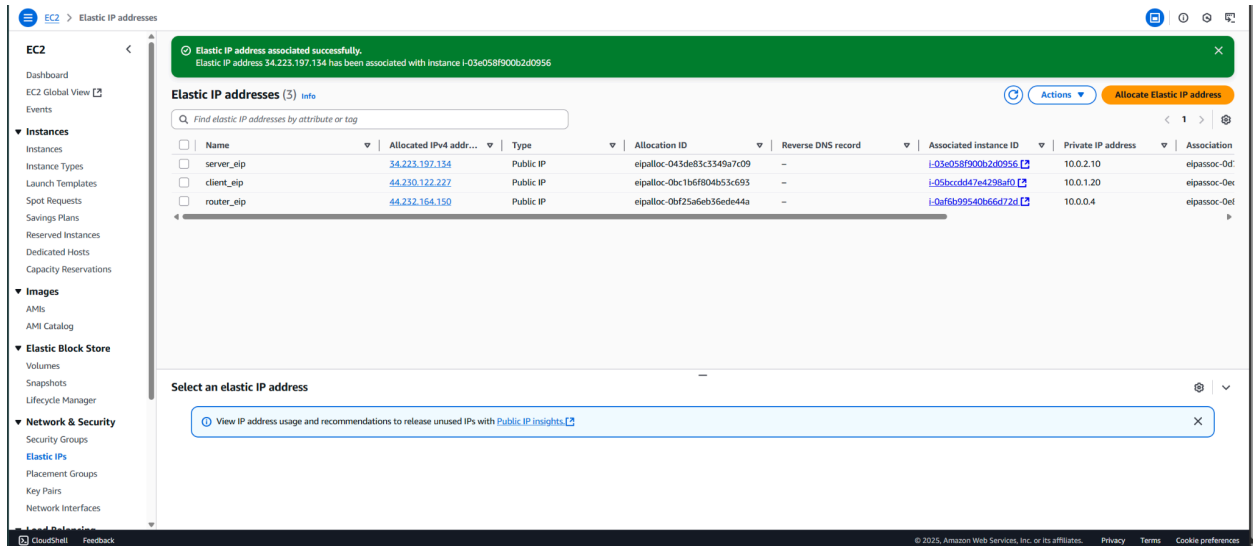
Launch instances

1

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6
☐	Lab2-Client-New	i-05bccdd47e4298af0	Running	t3.large	3/3 checks passed	View alarms	us-west-2a	-	44.230.122.227	44.230.122.227	-
☐	Lab2-Router	i-0af6b99540b66d72d	Running	t3.medium	3/3 checks passed	View alarms	us-west-2a	-	44.232.164.150	44.232.164.150	-
☐	Lab2-Server	i-03a058f900b2d0956	Running	t3.large	3/3 checks passed	View alarms	us-west-2a	-	34.223.197.134	34.223.197.134	-

Select an instance

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Routing & Firewall:

- Central router instance (Ubuntu instead of VyOS as using community AMIs) bridges the client and server subnets, with packet forwarding enabled and NAT for Internet access.
- Iptables used for forwarding and masquerade rules.
- Each VM has proper Security Group allowing ICMP, TCP/UDP (iperf), and SSH access.
- Route tables configured to ensure all traffic between client and server is routed via the router VM.

rtb-0977e61a185316661 / Router-RT

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0f9ba3abb216fce0b	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

rtb-0fe0fde1dab49d025 / Client-RT

Details Routes Subnet associations Edge associations Route propagation Tags

Routes (3) Both Edit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0f9ba3abb216fce0b	Active	No	Create Route
10.0.0.0/16	eni-0974ff34cd11c455a	Active	No	Create Route
10.0.2.0/24	eni-0974ff34cd11c455a	Active	No	Create Route

rtb-06b4a6b935fbc00ed / Server-RT

Details Routes Subnet associations Edge associations Route propagation Tags

Routes (3) Both Edit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0f9ba3abb216fce0b	Active	No	Create Route
10.0.0.0/16	eni-0ec3b7cc8a0432a76	Active	No	Create Route
10.0.1.0/24	eni-0ec3b7cc8a0432a76	Active	No	Create Route

Traceroute for Client VM and Server VM

```
root@ip-10-0-2-10:/home/ubuntu/Lab2# traceroute 10.0.1.20
traceroute to 10.0.1.20 (10.0.1.20), 30 hops max, 60 byte packets
 1 _gateway (10.0.2.1) 0.158 ms 0.169 ms 0.157 ms
 2 10.0.1.20 (10.0.1.20) 0.498 ms 0.545 ms 0.533 ms
```

```
root@ip-10-0-1-20:/home/ubuntu/Lab2# traceroute 10.0.2.10
traceroute to 10.0.2.10 (10.0.2.10), 30 hops max, 60 byte packets
 1 _gateway (10.0.1.1) 0.328 ms 0.297 ms 0.281 ms
 2 10.0.2.10 (10.0.2.10) 0.631 ms 0.753 ms 0.733 ms
root@ip-10-0-1-20:/home/ubuntu/Lab2#
```

Ip route on router:

```
root@ip-10-0-0-4:/home/ubuntu# ip route
default via 10.0.0.1 dev ens5 proto static
10.0.0.0/24 dev ens5 proto kernel scope link src 10.0.0.4
10.0.1.0/24 dev ens6 scope link
10.0.2.0/24 dev ens7 scope link
```

Case 1: Setting up the rate limiting to 100Mbps, RTT 10ms and 1% loss

Router:

```

root@ip-10-0-0-4:/home/ubuntu# tc qdisc show
qdisc noqueue 0: dev lo root refcnt 2
qdisc mq 0: dev ens5 root
qdisc fq_codel 0: dev ens5 parent :2 limit 10240p flows 1024 quantum 1514 target 5ms interval 100ms memory_limit 32Mb ecn drop_batch 64
qdisc fq_codel 0: dev ens6 parent :1 limit 10240p flows 1024 quantum 1514 target 5ms interval 100ms memory_limit 32Mb ecn drop_batch 64
qdisc tbf 1: dev ens6 root refcnt 3 rate 100Mbit burst 1250000b lat 0us
qdisc netem 10: dev ens6 parent 1:1 limit 1000 delay 5ms loss 1%
qdisc tbf 1: dev ens7 root refcnt 3 rate 100Mbit burst 1250000b lat 0us
qdisc netem 10: dev ens7 parent 1:1 limit 1000 delay 5ms loss 1%
root@ip-10-0-0-4:/home/ubuntu#

```

Client:

```

root@ip-10-0-1-20:/home/ubuntu# tc qdisc add dev ens5 root tbf rate 100mbit latency 0.001ms burst 1250000
root@ip-10-0-1-20:/home/ubuntu# traceroute 10.0.2.10
traceroute to 10.0.2.10 (10.0.2.10), 30 hops max, 60 byte packets
 1 _gateway (10.0.1.1) 5.348 ms 5.323 ms 5.310 ms
 2 10.0.2.10 (10.0.2.10) 10.588 ms 10.566 ms 10.552 ms
root@ip-10-0-1-20:/home/ubuntu# iperf -c 10.0.2.10

```

```

Client connecting to 10.0.2.10, TCP port 5001
TCP window size: 16.0 KByte (default)

```

```

[ 1] local 10.0.1.20 port 45356 connected with 10.0.2.10 port 5001 (icwnd/mss/irrt=14/1448/10613)
[ ID] Interval      Transfer      Bandwidth
[ 1] 0.0000-11.6079 sec 32.6 MBytes  23.6 Mbits/sec
root@ip-10-0-1-20:/home/ubuntu# iperf -c 10.0.2.10 -u -b 200mbit

```

```

Client connecting to 10.0.2.10, UDP port 5001
Sending 1470 byte datagrams, IPG target: 58.80 us (kalman adjust)
UDP buffer size: 208 KByte (default)

```

```

[ 1] local 10.0.1.20 port 47004 connected with 10.0.2.10 port 5001
[ ID] Interval      Transfer      Bandwidth      Jitter    Lost/Total Datagrams
[ 1] 0.0000-10.0007 sec 117 MBytes  98.2 Mbits/sec
[ 1] Sent 83543 datagrams
[ 1] Server Report:
[ ID] Interval      Transfer      Bandwidth      Jitter    Lost/Total Datagrams
[ 1] 0.0000-10.0063 sec 116 MBytes  97.2 Mbits/sec  0.232 ms 841/83542 (1%)
root@ip-10-0-1-20:/home/ubuntu#

```

```

root@ip-10-0-1-20:/home/ubuntu# tc qdisc add dev ens5 root tbf rate 100mbit latency 0.001ms burst 1250000
root@ip-10-0-1-20:/home/ubuntu# traceroute 10.0.2.10
traceroute to 10.0.2.10 (10.0.2.10), 30 hops max, 60 byte packets
 1 _gateway (10.0.1.1) 5.348 ms 5.323 ms 5.310 ms
 2 10.0.2.10 (10.0.2.10) 10.588 ms 10.566 ms 10.552 ms
root@ip-10-0-1-20:/home/ubuntu# iperf -c 10.0.2.10

```

```

Client connecting to 10.0.2.10, TCP port 5001
TCP window size: 16.0 KByte (default)

```

```

[ 1] local 10.0.1.20 port 45356 connected with 10.0.2.10 port 5001 (icwnd/mss/irrt=14/1448/10613)
[ ID] Interval      Transfer      Bandwidth
[ 1] 0.0000-11.6079 sec 32.6 MBytes  23.6 Mbits/sec
root@ip-10-0-1-20:/home/ubuntu# iperf -c 10.0.2.10 -u -b 200mbit

```

```

Client connecting to 10.0.2.10, UDP port 5001
Sending 1470 byte datagrams, IPG target: 58.80 us (kalman adjust)
UDP buffer size: 208 KByte (default)

```

```

[ 1] local 10.0.1.20 port 47004 connected with 10.0.2.10 port 5001
[ ID] Interval      Transfer      Bandwidth      Jitter    Lost/Total Datagrams
[ 1] 0.0000-10.0007 sec 117 MBytes  98.2 Mbits/sec
[ 1] Sent 83543 datagrams
[ 1] Server Report:
[ ID] Interval      Transfer      Bandwidth      Jitter    Lost/Total Datagrams
[ 1] 0.0000-10.0063 sec 116 MBytes  97.2 Mbits/sec  0.232 ms 841/83542 (1%)
root@ip-10-0-1-20:/home/ubuntu#

```

```

root@ip-10-0-1-20:/home/ubuntu/Lab2# iperf -c 10.0.2.10
-----
Client connecting to 10.0.2.10, TCP port 5001
TCP window size: 16.0 KByte (default)
-----
[ 1] local 10.0.1.20 port 56802 connected with 10.0.2.10 port 5001 (icwnd/mss/irrt=87/8948/10787)
[ ID] Interval      Transfer      Bandwidth
[ 1] 0.0000-10.2363 sec  109 MBytes   89.6 Mbits/sec
root@ip-10-0-1-20:/home/ubuntu/Lab2#

```

Server:

```

root@ip-10-0-2-10:/home/ubuntu# tc qdisc add dev ens5 root tbf rate 100mbit latency 0.001ms burst 1250000
root@ip-10-0-2-10:/home/ubuntu# traceroute 10.0.1.20
traceroute to 10.0.1.20 (10.0.1.20), 30 hops max, 60 byte packets
 1 _gateway (10.0.2.1)  5.256 ms  5.233 ms  5.211 ms
 2 10.0.1.20 (10.0.1.20) 10.590 ms 10.574 ms *
root@ip-10-0-2-10:/home/ubuntu# iperf -s

```

```

-----
Server listening on TCP port 5001
TCP window size: 128 KByte (default)
-----
[ 1] local 10.0.2.10 port 5001 connected with 10.0.1.20 port 45356 (icwnd/mss/irrt=14/1448/10549)
[ ID] Interval      Transfer      Bandwidth
[ 1] 0.0000-11.5955 sec  32.6 MBytes   23.6 Mbits/sec
^Croot@ip-10-0-2-10:/home/ubuntu# iperf -s -u

```

```

-----
Server listening on UDP port 5001
UDP buffer size: 208 KByte (default)
-----
[ 1] local 10.0.2.10 port 5001 connected with 10.0.1.20 port 47004
[ ID] Interval      Transfer      Bandwidth      Jitter    Lost/Total Datagrams
[ 1] 0.0000-10.0063 sec  116 MBytes   97.2 Mbits/sec  0.233 ms  841/83542 (1%)
[ 3] WARNING: ack of last datagram failed.

```

```

root@ip-10-0-2-10:/home/ubuntu# tc qdisc add dev ens5 root tbf rate 100mbit latency 0.001ms burst 1250000
root@ip-10-0-2-10:/home/ubuntu# traceroute 10.0.1.20
traceroute to 10.0.1.20 (10.0.1.20), 30 hops max, 60 byte packets
 1 _gateway (10.0.2.1)  5.256 ms  5.233 ms  5.211 ms
 2 10.0.1.20 (10.0.1.20) 10.590 ms 10.574 ms *
root@ip-10-0-2-10:/home/ubuntu# iperf -s

```

```

-----
Server listening on TCP port 5001
TCP window size: 128 KByte (default)
-----
[ 1] local 10.0.2.10 port 5001 connected with 10.0.1.20 port 45356 (icwnd/mss/irrt=14/1448/10549)
[ ID] Interval      Transfer      Bandwidth
[ 1] 0.0000-11.5955 sec  32.6 MBytes   23.6 Mbits/sec
^Croot@ip-10-0-2-10:/home/ubuntu# iperf -s -u

```

```

-----
Server listening on UDP port 5001
UDP buffer size: 208 KByte (default)
-----
[ 1] local 10.0.2.10 port 5001 connected with 10.0.1.20 port 47004
[ ID] Interval      Transfer      Bandwidth      Jitter    Lost/Total Datagrams
[ 1] 0.0000-10.0063 sec  116 MBytes   97.2 Mbits/sec  0.233 ms  841/83542 (1%)
[ 3] WARNING: ack of last datagram failed.

```

```

root@ip-10-0-2-10:/home/ubuntu/Lab2# iperf -s
-----
Server listening on TCP port 5001
TCP window size: 128 KByte (default)
-----
[ 1] local 10.0.2.10 port 5001 connected with 10.0.1.20 port 56802 (icwnd/mss/irrt=87/8948/10618)
[ ID] Interval      Transfer      Bandwidth
[ 1] 0.0000-10.2205 sec  109 MBytes   89.8 Mbits/sec

```

Testing the file transfer for case 1, MTU 1500:

Client

```
root@ip-10-0-1-20:/home/ubuntu/Lab2# ip link show ens5
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc tbf state UP mode DEFAULT group default qlen 1000
    link/ether 06:21:21:d2:9a:67 brd ff:ff:ff:ff:ff:ff
    altname enp0s5
root@ip-10-0-1-20:/home/ubuntu/Lab2# ./ft_client --verbose --chunk 1460 --batch 32 data.bin 10.0.2.10 5510
chunk_bytes=1460
Transfer size = 8589.93 Mbits
Elapsed time = 98.1701 sec
Throughput = 87.5005 Mbit/s
root@ip-10-0-1-20:/home/ubuntu/Lab2# md5sum data.bin
5d658943d977b6f29aae2f4950dcf232 data.bin
root@ip-10-0-1-20:/home/ubuntu/Lab2#
```

Server:

```
root@ip-10-0-2-10:/home/ubuntu/Lab2# ip link show ens5
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc tbf state UP mode DEFAULT group default qlen 1000
    link/ether 06:fa:3f:5f:07:41 brd ff:ff:ff:ff:ff:ff
    altname enp0s5
root@ip-10-0-2-10:/home/ubuntu/Lab2# ./ft_server --verbose out.bin 5510
^C
root@ip-10-0-2-10:/home/ubuntu/Lab2# ./ft_server out.bin 5510
File transmission complete.
root@ip-10-0-2-10:/home/ubuntu/Lab2# md5sum out.bin
5d658943d977b6f29aae2f4950dcf232 out.bin
root@ip-10-0-2-10:/home/ubuntu/Lab2#
```

Throughput: 87.5005 Mbit/s

Testing the file transfer for case 1, MTU 9000:

Client:

```
root@ip-10-0-1-20:/home/ubuntu/Lab2# ip link set dev ens5 mtu 9000 up
ip link show ens5
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9000 qdisc tbf state UP mode DEFAULT group default qlen 1000
    link/ether 06:21:21:d2:9a:67 brd ff:ff:ff:ff:ff:ff
    altname enp0s5
root@ip-10-0-1-20:/home/ubuntu/Lab2# traceroute 10.0.2.10
1  _gateway (10.0.1.1)  5.634 ms  5.460 ms  5.438 ms
2  10.0.2.10 (10.0.2.10)  10.753 ms  10.717 ms  10.702 ms
root@ip-10-0-1-20:/home/ubuntu/Lab2# ./ft_client --verbose --chunk 8960 --batch 32 data.bin 10.0.2.10 5510
chunk_bytes=8960
Transfer size = 8589.93 Mbits
Elapsed time = 98.5062 sec
Throughput = 87.202 Mbit/s
root@ip-10-0-1-20:/home/ubuntu/Lab2# md5sum data.bin
5d658943d977b6f29aae2f4950dcf232 data.bin
root@ip-10-0-1-20:/home/ubuntu/Lab2#
```

Server:

```

root@ip-10-0-2-10:/home/ubuntu/Lab2# ip link set dev ens5 mtu 9000 up
ip link show ens5
traceroute 10.0.1.20
./ft_server out.bin 5510
md5sum out.bin
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9000 qdisc tbf state UP mode DEFAULT group default qlen 1000
    link/ether 06:fa:3f:5f:07:41 brd ff:ff:ff:ff:ff:ff
    altnam enp0s5
traceroute to 10.0.1.20 (10.0.1.20), 30 hops max, 60 byte packets
 1  _gateway (10.0.2.1)  5.243 ms  5.213 ms  5.182 ms
 2  10.0.1.20 (10.0.1.20) 10.562 ms 10.543 ms 10.523 ms
File transmission complete.
5d658943d977b6f29aae2f4950dcf232 out.bin
root@ip-10-0-2-10:/home/ubuntu/Lab2#

```

Throughput: 87.202 Mbit/s

Case 2: Setting up the RTT 200ms and 20% loss

Router:

```

root@ip-10-0-1-20:/home/ubuntu# tc qdisc show
qdisc noqueue 0: dev lo root refcnt 2
qdisc mq 0: dev ens5 root
qdisc fq_codel 0: dev ens5 parent :2 limit 10240p flows 1024 quantum 1514 target 5ms interval 100ms memory_limit 32Mb ecn drop_batch 64
qdisc fq_codel 0: dev ens5 parent :1 limit 10240p flows 1024 quantum 1514 target 5ms interval 100ms memory_limit 32Mb ecn drop_batch 64
qdisc tbf 1: dev ens6 root refcnt 3 rate 100Mbit burst 1250000b lat 0us
qdisc netem 10: dev ens6 parent 1:1 limit 1000 delay 100ms loss 20%
qdisc tbf 1: dev ens7 root refcnt 3 rate 100Mbit burst 1250000b lat 0us
qdisc netem 10: dev ens7 parent 1:1 limit 1000 delay 100ms loss 20%

```

Client:

```

--- 10.0.2.10 ping statistics ---
200 packets transmitted, 141 received, 29.5% packet loss, time 40285ms
rtt min/avg/max/mdev = 200.528/200.653/200.985/0.073 ms, pipe 2
root@ip-10-0-1-20:/home/ubuntu/Lab2#

```

```

root@ip-10-0-1-20:/home/ubuntu/Lab2# traceroute 10.0.2.10
traceroute to 10.0.2.10 (10.0.2.10), 30 hops max, 60 byte packets
 1  _gateway (10.0.1.1) 100.373 ms * 100.324 ms
 2  10.0.2.10 (10.0.2.10) 200.655 ms 200.628 ms 200.613 ms
root@ip-10-0-1-20:/home/ubuntu/Lab2#

```

Server:

```

64 bytes from 10.0.1.20: icmp_seq=198 ttl=63 time=201 ms
64 bytes from 10.0.1.20: icmp_seq=199 ttl=63 time=201 ms

--- 10.0.1.20 ping statistics ---
200 packets transmitted, 125 received, 37.5% packet loss, time 40396ms
rtt min/avg/max/mdev = 200.492/200.609/201.043/0.070 ms

```

```

root@ip-10-0-2-10:/home/ubuntu/Lab2# traceroute 10.0.1.20
traceroute to 10.0.1.20 (10.0.1.20), 30 hops max, 60 byte packets
 1  _gateway (10.0.2.1) 100.248 ms 100.227 ms 100.215 ms
 2  10.0.1.20 (10.0.1.20) 200.603 ms 200.576 ms 200.563 ms

```

Iperf tests:

Server:

```
root@ip-10-0-2-10:/home/ubuntu/Lab2# iperf -s -u
-----
Server listening on UDP port 5001
UDP buffer size: 208 KByte (default)
-----
[ 1] local 10.0.2.10 port 5001 connected with 10.0.1.20 port 39094
[ ID] Interval      Transfer    Bandwidth      Jitter    Lost/Total Datagrams
[ 1] 0.0000-10.0107 sec 93.2 MBytes 78.1 Mbits/sec  0.286 ms 17134/83581 (20%)
[ 3] WARNING: ack of last datagram failed.

root@ip-10-0-2-10:/home/ubuntu/Lab2# iperf -s
-----
Server listening on TCP port 5001
TCP window size: 128 KByte (default)
-----
[ 1] local 10.0.2.10 port 5001 connected with 10.0.1.20 port 49402 (icwnd/mss/irrt=14/1448/201121)
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.0000-26.8100 sec 260 KBytes 79.5 Kbits/sec
```

Client:

```
root@ip-10-0-1-20:/home/ubuntu/Lab2# iperf -c 10.0.2.10 -u -b 200mbit
-----
Client connecting to 10.0.2.10, UDP port 5001
Sending 1470 byte datagrams, IPG target: 58.80 us (kalman adjust)
UDP buffer size: 208 KByte (default)
-----
[ 1] local 10.0.1.20 port 39094 connected with 10.0.2.10 port 5001
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.0000-10.0051 sec 117 MBytes 98.2 Mbits/sec
[ 1] Sent 83582 datagrams
[ 1] Server Report:
[ ID] Interval      Transfer    Bandwidth      Jitter    Lost/Total Datagrams
[ 1] 0.0000-10.0107 sec 93.2 MBytes 78.1 Mbits/sec  0.286 ms 17134/83581 (20%)

root@ip-10-0-1-20:/home/ubuntu/Lab2# iperf -c 10.0.2.10
-----
Client connecting to 10.0.2.10, TCP port 5001
TCP window size: 16.0 KByte (default)
-----
[ 1] local 10.0.1.20 port 49402 connected with 10.0.2.10 port 5001 (icwnd/mss/irrt=14/1448/200000)
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.0000-20.7270 sec 260 KBytes 103 Kbits/sec
```

Testing the file transfer for case 2, MTU 1500:

Client:


```

root@ip-10-0-1-20:/home/ubuntu/Lab2# ip link show ens5
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc tbf state UP mode DEFAULT group default qlen 1000
    link/ether 06:21:21:d2:9a:67 brd ff:ff:ff:ff:ff:ff
    altname enp0s5
root@ip-10-0-1-20:/home/ubuntu/Lab2# ./ft_client --verbose --chunk 1460 --batch 32 data.bin 10.0.2.10 5510
chunk_bytes=1460
Transfer size = 8589.93 Mbits
Elapsed time = 144.442 sec
Throughput = 59.4697 Mbit/s
root@ip-10-0-1-20:/home/ubuntu/Lab2# md5sum data.bin
5d658943d977b6f29aae2f4950dcf232 data.bin
root@ip-10-0-1-20:/home/ubuntu/Lab2# traceroute 10.0.2.10
traceroute to 10.0.2.10 (10.0.2.10), 30 hops max, 60 byte packets
 1 * _gateway (10.0.1.1) 100.891 ms 100.877 ms
 2 10.0.2.10 (10.0.2.10) 200.559 ms * *
root@ip-10-0-1-20:/home/ubuntu/Lab2# _

```

Server:

```

root@ip-10-0-2-10:/home/ubuntu/Lab2# ip link show ens5
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc tbf state UP mode DEFAULT group default qlen 1000
    link/ether 06:fa:3f:5f:07:41 brd ff:ff:ff:ff:ff:ff
    altname enp0s5
root@ip-10-0-2-10:/home/ubuntu/Lab2# ./ft_server out.bin 5510
File transmission complete.
root@ip-10-0-2-10:/home/ubuntu/Lab2# md5sum out.bin
5d658943d977b6f29aae2f4950dcf232 out.bin
root@ip-10-0-2-10:/home/ubuntu/Lab2# traceroute 10.0.1.20
traceroute to 10.0.1.20 (10.0.1.20), 30 hops max, 60 byte packets
 1 _gateway (10.0.2.1) 100.273 ms 100.250 ms 100.239 ms
 2 * 10.0.1.20 (10.0.1.20) 200.790 ms 200.775 ms
root@ip-10-0-2-10:/home/ubuntu/Lab2# _

```

Throughput: 59.4697 Mbit/s

Testing the file transfer for case 2, MTU 9000:

Client:

```

root@ip-10-0-1-20:/home/ubuntu/Lab2# ip link set dev ens5 mtu 9000 up
ip link show ens5
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9000 qdisc tbf state UP mode DEFAULT group default qlen 1000
    link/ether 06:21:21:d2:9a:67 brd ff:ff:ff:ff:ff:ff
    altname enp0s5
root@ip-10-0-1-20:/home/ubuntu/Lab2# traceroute 10.0.2.10
./ft_client --verbose --chunk 8960 --batch 32 data.bin 10.0.2.10 5510
md5sum out.bin
5d658943d977b6f29aae2f4950dcf232 out.bin
root@ip-10-0-1-20:/home/ubuntu/Lab2# traceroute 10.0.2.10
traceroute to 10.0.2.10 (10.0.2.10), 30 hops max, 60 byte packets
 1 _gateway (10.0.1.1) 100.561 ms * *
 2 10.0.2.10 (10.0.2.10) 200.709 ms 200.684 ms 200.667 ms
chunk_bytes=8960
Transfer size = 8589.93 Mbits
Elapsed time = 195.803 sec
Throughput = 43.8702 Mbit/s
md5sum: out.bin: No such file or directory
root@ip-10-0-1-20:/home/ubuntu/Lab2# md5sum data.bin
5d658943d977b6f29aae2f4950dcf232 data.bin
root@ip-10-0-1-20:/home/ubuntu/Lab2# _

```

Server:

```

root@ip-10-0-2-10:/home/ubuntu/Lab2# ip link set dev ens5 mtu 9000 up
ip link show ens5
traceroute 10.0.1.20
./ft_server out.bin 5510
md5sum out.bin
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9000 qdisc tbf state UP mode DEFAULT group default qlen 1000
    link/ether 06:fa:3f:5f:07:41 brd ff:ff:ff:ff:ff:ff
    altname enp0s5
traceroute to 10.0.1.20 (10.0.1.20), 30 hops max, 60 byte packets
 1 * _gateway (10.0.2.1) 100.272 ms 100.259 ms
 2 10.0.1.20 (10.0.1.20) 200.649 ms * *
File transmission complete.
5d658943d977b6f29aae2f4950dcf232 out.bin
root@ip-10-0-2-10:/home/ubuntu/Lab2#

```

Throughput: 43.8702 Mbit/s

Case 3: Setting up the RTT of 200ms, No packet loss, but the network transfer rate is reduced by 20%

Router:

```

root@ip-10-0-0-4:/home/ubuntu# tc qdisc show
qdisc noqueue 0: dev lo root refcnt 2
qdisc mq 0: dev ens5 root
qdisc fq_codel 0: dev ens5 parent :2 limit 10240p flows 1024 quantum 1514 target 5ms interval 100ms memory_limit 32Mb ecn drop_batch 64
qdisc fq_codel 0: dev ens5 parent :1 limit 10240p flows 1024 quantum 1514 target 5ms interval 100ms memory_limit 32Mb ecn drop_batch 64
qdisc tbf 1: dev ens6 root refcnt 3 rate 80Mbit burst 1250000b lat 1us
qdisc netem 10: dev ens6 parent 1:1 limit 1000 delay 100ms
qdisc tbf 1: dev ens7 root refcnt 3 rate 80Mbit burst 1250000b lat 1us
qdisc netem 10: dev ens7 parent 1:1 limit 1000 delay 100ms

```

Client:

```

--- 10.0.2.10 ping statistics ---
200 packets transmitted, 200 received, 0% packet loss, time 39933ms
rtt min/avg/max/mdev = 200.464/200.566/200.944/0.064 ms, pipe 2
root@ip-10-0-1-20:/home/ubuntu/Lab2#

```

```

root@ip-10-0-2-10:/home/ubuntu/Lab2# traceroute 10.0.1.20
traceroute to 10.0.1.20 (10.0.1.20), 30 hops max, 60 byte packets
 1 _gateway (10.0.2.1) 100.450 ms 100.419 ms 100.403 ms
 2 10.0.1.20 (10.0.1.20) 200.740 ms 200.708 ms *

```

Server:

```

--- 10.0.1.20 ping statistics ---
200 packets transmitted, 200 received, 0% packet loss, time 39933ms
rtt min/avg/max/mdev = 200.483/200.577/200.869/0.067 ms, pipe 2
root@ip-10-0-2-10:/home/ubuntu/Lab2#

```

```

root@ip-10-0-1-20:/home/ubuntu/Lab2# traceroute 10.0.2.10
traceroute to 10.0.2.10 (10.0.2.10), 30 hops max, 60 byte packets
 1 _gateway (10.0.1.1) 100.360 ms 100.339 ms 100.328 ms
 2 10.0.2.10 (10.0.2.10) 200.574 ms 200.560 ms 200.549 ms

```

Iperf tests:

Server:

```
root@ip-10-0-2-10:/home/ubuntu/Lab2# iperf -s -u
-----
Server listening on UDP port 5001
UDP buffer size: 208 KByte (default)
-----
[ 1] local 10.0.2.10 port 5001 connected with 10.0.1.20 port 44232
[ ID] Interval      Transfer    Bandwidth      Jitter    Lost/Total Datagrams
[ 1] 0.0000-10.0607 sec 94.4 MBytes 78.7 Mbits/sec  0.220 ms 16203/83564 (19%)
[ 3] WARNING: ack of last datagram failed.

^Croot@ip-10-0-2-10:/home/ubuntu/Lab2# iperf -s
-----
Server listening on TCP port 5001
TCP window size: 128 KByte (default)
-----
[ 1] local 10.0.2.10 port 5001 connected with 10.0.1.20 port 57712 (icwnd/mss/irrt=87/8948/200736)
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.0000-10.1404 sec 88.5 MBytes 73.2 Mbits/sec
```

Client:

```
root@ip-10-0-1-20:/home/ubuntu/Lab2# iperf -u -c 10.0.2.10 -b 100mbit
-----
Client connecting to 10.0.2.10, UDP port 5001
Sending 1470 byte datagrams, IPG target: 117.60 us (kalman adjust)
UDP buffer size: 208 KByte (default)
-----
[ 1] local 10.0.1.20 port 44232 connected with 10.0.2.10 port 5001
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.0000-10.0036 sec 117 MBytes 98.2 Mbits/sec
[ 1] Sent 83565 datagrams
[ 1] Server Report:
[ ID] Interval      Transfer    Bandwidth      Jitter    Lost/Total Datagrams
[ 1] 0.0000-10.0607 sec 94.4 MBytes 78.7 Mbits/sec  0.219 ms 16203/83564 (19%)
root@ip-10-0-1-20:/home/ubuntu/Lab2#

root@ip-10-0-1-20:/home/ubuntu/Lab2# iperf -c 10.0.2.10
-----
Client connecting to 10.0.2.10, TCP port 5001
TCP window size: 16.0 KByte (default)
-----
[ 1] local 10.0.1.20 port 57712 connected with 10.0.2.10 port 5001 (icwnd/mss/irrt=87/8948/200760)
[ ID] Interval      Transfer    Bandwidth
[ 1] 0.0000-10.3562 sec 88.5 MBytes 71.7 Mbits/sec
root@ip-10-0-1-20:/home/ubuntu/Lab2#
```

Though the TCP performance is

- Modern Congestion Control: Algorithms like CUBIC/BBR recover from losses faster than classical TCP
- Large TCP Windows: Modern default buffers allow high bandwidth×delay product

- Fast Recovery: Advanced loss detection and recovery mechanisms
- Selective Acknowledgment (SACK): Only retransmits lost segments, not entire windows

```
^Croot@ip-10-0-2-10:/home/ubuntu/Lab2# sysctl net.ipv4.tcp_congestion_control
net.ipv4.tcp_congestion_control = cubic
root@ip-10-0-2-10:/home/ubuntu/Lab2# sysctl net.core.rmem_max net.core.wmem_max
sysctl net.ipv4.tcp_rmem net.ipv4.tcp_wmem
net.core.rmem_max = 212992
net.core.wmem_max = 212992
net.ipv4.tcp_rmem = 4096          131072   6291456
net.ipv4.tcp_wmem = 4096          16384   4194304
```

```
root@ip-10-0-1-20:/home/ubuntu/Lab2# sysctl net.ipv4.tcp_congestion_control
net.ipv4.tcp_congestion_control = cubic
root@ip-10-0-1-20:/home/ubuntu/Lab2# sysctl net.core.rmem_max net.core.wmem_max
sysctl net.ipv4.tcp_rmem net.ipv4.tcp_wmem
net.core.rmem_max = 212992
net.core.wmem_max = 212992
net.ipv4.tcp_rmem = 4096          131072   6291456
net.ipv4.tcp_wmem = 4096          16384   4194304
```

Testing the file transfer for case 3, MTU 1500:

Client:

```
root@ip-10-0-1-20:/home/ubuntu/Lab2# ip link set dev ens5 mtu 1500 up
ip link show ens5
traceroute 10.0.2.10
./ft_client --verbose --chunk 1460 --batch 32 data.bin 10.0.2.10 5510
md5sum out.bin
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc tbf state UP mode DEFAULT group default qlen 1000
    link/ether 06:21:21:d2:9a:67 brd ff:ff:ff:ff:ff:ff
    altname enp0s5
traceroute to 10.0.2.10 (10.0.2.10), 30 hops max, 60 byte packets
 1 _gateway (10.0.1.1) 100.369 ms 100.337 ms 100.319 ms
 2 10.0.2.10 (10.0.2.10) 200.714 ms 200.694 ms *
chunk_bytes=1460
Transfer size = 8589.93 Mbits
Elapsed time = 147.588 sec
Throughput = 58.2019 Mbit/s
md5sum: out.bin: No such file or directory
root@ip-10-0-1-20:/home/ubuntu/Lab2# md5sum data.bin
5d658943d977b6f29aae2f4950dcf232 data.bin
root@ip-10-0-1-20:/home/ubuntu/Lab2#
```

Server:

```

root@ip-10-0-2-10:/home/ubuntu/Lab2# ip link set dev ens5 mtu 1500 up
ip link show ens5
traceroute 10.0.1.20
./ft_server out.bin 5510
md5sum out.bin
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc tbf state UP mode DEFAULT group default qlen 1000
    link/ether 06:fa:3f:5f:07:41 brd ff:ff:ff:ff:ff:ff
    altname enp0s5
traceroute to 10.0.1.20 (10.0.1.20), 30 hops max, 60 byte packets
 1 _gateway (10.0.2.1) 100.351 ms 100.328 ms 100.304 ms
 2 10.0.1.20 (10.0.1.20) 200.622 ms 200.602 ms 200.588 ms
File transmission complete.
5d658943d977b6f29aae2f4950dcf232 out.bin
root@ip-10-0-2-10:/home/ubuntu/Lab2#

```

Throughput: 58.2019 Mbit/s

Testing the file transfer for case 3, MTU 9000:

Client:

```

root@ip-10-0-1-20:/home/ubuntu/Lab2# ip link set dev ens5 mtu 9000 up
ip link show ens5
traceroute 10.0.2.10
./ft_client --verbose --chunk 8960 --batch 32 data.bin 10.0.2.10 5510
md5sum data.bin
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9000 qdisc mq state UP mode DEFAULT group default qlen 1000
    link/ether 06:21:21:d2:9a:67 brd ff:ff:ff:ff:ff:ff
    altname enp0s5
traceroute to 10.0.2.10 (10.0.2.10), 30 hops max, 60 byte packets
 1 _gateway (10.0.1.1) 100.400 ms 100.366 ms 100.349 ms
 2 10.0.2.10 (10.0.2.10) 200.587 ms 200.567 ms 200.547 ms
chunk_bytes=8960
Transfer size = 8589.93 Mbits
Elapsed time = 127.189 sec
Throughput = 67.5369 Mbit/s
5d658943d977b6f29aae2f4950dcf232 data.bin
root@ip-10-0-1-20:/home/ubuntu/Lab2#

```

Server:

```

root@ip-10-0-2-10:/home/ubuntu/Lab2# ip link set dev ens5 mtu 9000 up
ip link show ens5
traceroute 10.0.1.20
./ft_server out.bin 5510
md5sum out.bin
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9000 qdisc mq state UP mode DEFAULT group default qlen 1000
    link/ether 06:fa:3f:5f:07:41 brd ff:ff:ff:ff:ff:ff
    altname enp0s5
traceroute to 10.0.1.20 (10.0.1.20), 30 hops max, 60 byte packets
 1 _gateway (10.0.2.1) 100.258 ms 100.233 ms 100.222 ms
 2 10.0.1.20 (10.0.1.20) 200.831 ms 200.806 ms 200.853 ms
File transmission complete.
5d658943d977b6f29aae2f4950dcf232 out.bin
root@ip-10-0-2-10:/home/ubuntu/Lab2#

```

Throughput: 67.5369 Mbit/s

